

Hw09 - Universal Kriging/Autoregressive Homework

Problem 1 (Autoregressive modeling)

The file `penn_spdf.rda` contains an `sf data.frame` related to lung cancer deaths for 67 Pennsylvania counties in 2002. The data are simplified from the `pennLC` data available in the **SpatialEpi** package. The data frame includes the variables:

- `cases` : the number of lung cancer cases in the county
- `pop` : the county population of the counties
- `rate` : the lung cancer rate (`cases / pop`)
- `smoking` : the proportion of the population that smokes.

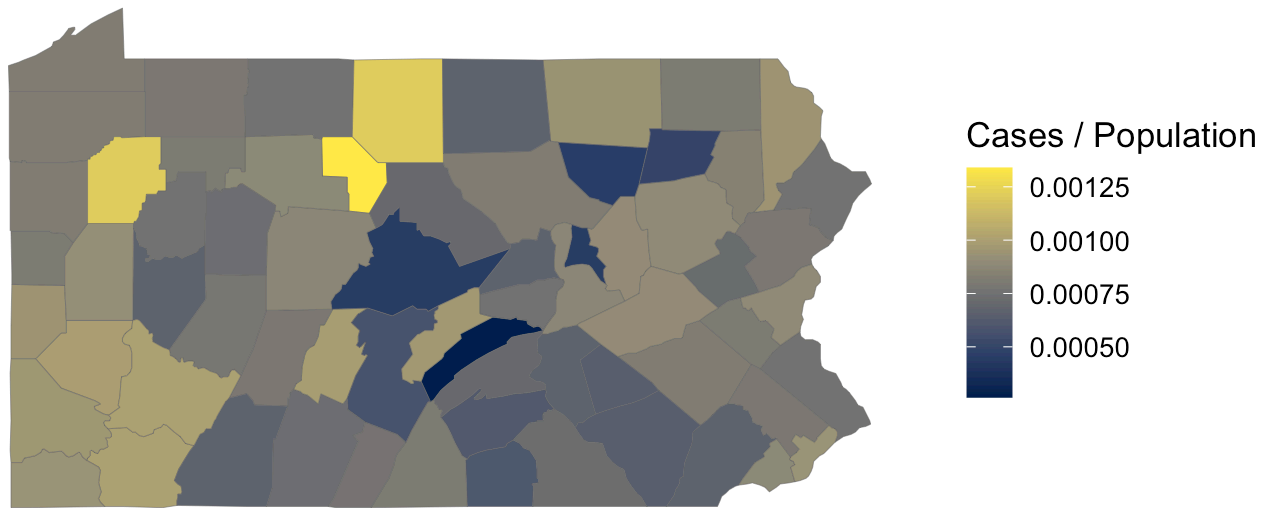
(a)

Create a choropleth map of lung cancer `rate`.

Solution

```
ggplot(penn_sdf) +  
  geom_sf(aes(fill = rate), color = "gray40", linewidth = 0.1) +  
  scale_fill_viridis_c(option = "cividis", na.value = "white") +  
  labs(  
    title = "Lung Cancer Rate",  
    fill = "Cases / Population"  
  ) +  
  theme_minimal(base_size = 13) +  
  theme(  
    axis.text = element_blank(),  
    axis.title = element_blank(),  
    panel.grid = element_blank()  
  )
```

Lung Cancer Rate



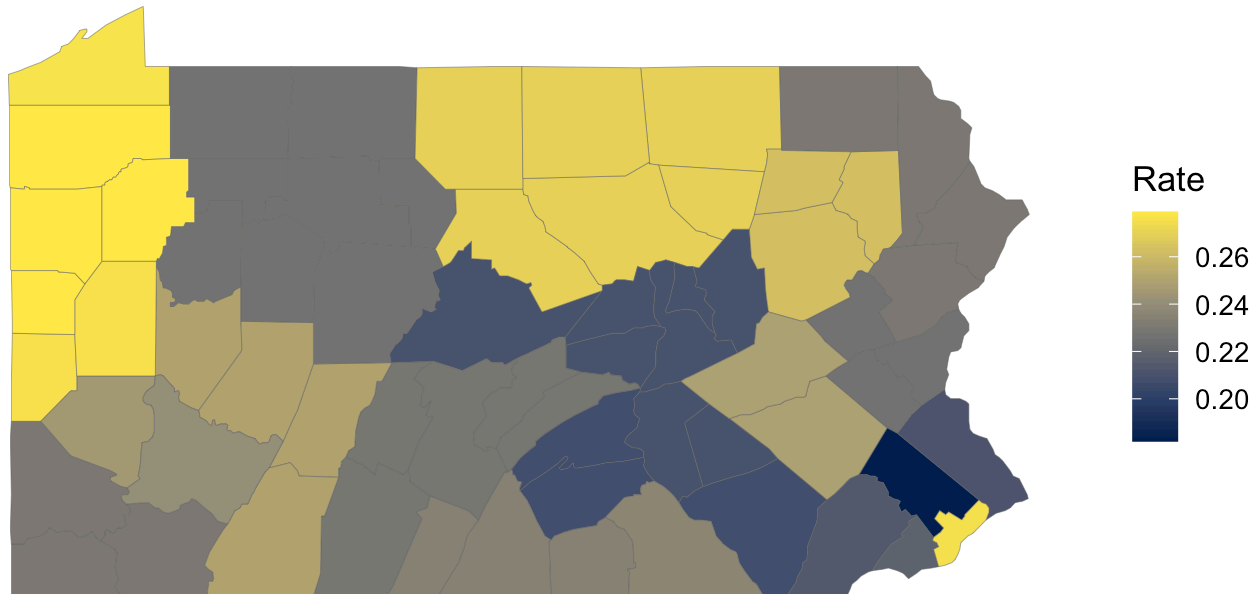
(b)

Create a choropleth map of `smoking`.

Solution

```
ggplot(penn_sfdf) +  
  geom_sf(aes(fill = smoking), color = "gray40", linewidth = 0.1) +  
  scale_fill_viridis_c(option = "cividis", na.value = "white") +  
  labs(  
    title = "Smoking Proportion",  
    fill = "Rate"  
  ) +  
  theme_minimal(base_size = 13) +  
  theme(  
    axis.text = element_blank(),  
    axis.title = element_blank(),  
    panel.grid = element_blank()  
  )
```

Smoking Proportion



(c)

Does there appear to be evidence of clustering of lung cancer rate/smoking in the individual maps? Are there similar patterns in the two maps?

Solution

There does seem to be some evidence of clustering for smoking population as well as slight clustering for cancer however this is a little more spread out. As, we can see however it does seem like one of the northern counties with high smoking rate also has high rates of cancer. However, they do not seem to be completely similar it is very unclear.

(d)

Describe what is occurring in the code below.

```
library(spdep)
pennnb = spdep::poly2nb(penn_sfd)
pennwb <- nb2listw(pennnb, style = "B")
pennws <- nb2listw(pennnb, style = "W")
```

Solution

The `poly2nb` gives us a neighbours list based on regions with continuous boundaries. So it gives us a list of neighboring counties for the `penn_sfdf` for each of the counties. It then creates two sets of weights, one which is in binaries and the other which gives us row standardized weights.

(e)

The code below runs 8 different linear model-based Moran's I tests for global spatial autocorrelation. The fitted linear models used include:

- no covariates (just an intercept)
- no covariates while adjusting the residual weights to be related to population size.
- the smoking covariate
- the smoking covariate while adjusting the residual weights to be related to population size.

The Moran's I test was run for each model with both binary and row-standardized spatial weights matrices. The results are summarized in the table below.

covariates	residual weights	spatial weights
none	equal	binary
none	pop	binary
smoking	equal	binary
smoking	pop	binary
none	equal	row-standardized
none	pop	row-standardized
smoking	equal	row-standardized
smoking	pop	row-standardized

Does weighting the residuals affect the results? Do the covariates included affect the results? Does the spatial weights matrices used affect the results?

```
# evaluate evidence for spatial dependence based on whether
# covariates are included and heterogeneous weights are accounted for
pennlm1 = lm(rate ~ 1, data = penn_sfdf)
pennlm2 = lm(rate ~ 1, data = penn_sfdf, weights = pop)
pennlm3 = lm(rate ~ smoking, data = penn_sfdf)
pennlm4 = lm(rate ~ smoking, data = penn_sfdf, weights = pop)

lm.morantest(pennlm1, pennwb)
```

Global Moran I for regression residuals

data:

model: `lm(formula = rate ~ 1, data = penn_sfdf)`weights: `pennwb`

Moran I statistic standard deviate = 1.2284, p-value = 0.1096

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.073813364	-0.015151515	0.005244855

`lm.morantest(pennlm2, pennwb)`

Global Moran I for regression residuals

data:

model: `lm(formula = rate ~ 1, data = penn_sfdf, weights = pop)`weights: `pennwb`

Moran I statistic standard deviate = 3.3737, p-value = 0.0003709

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.233659603	-0.012654910	0.005330586

`lm.morantest(pennlm3, pennwb)`

Global Moran I for regression residuals

data:

model: `lm(formula = rate ~ smoking, data = penn_sfdf)`weights: `pennwb`

Moran I statistic standard deviate = 0.64821, p-value = 0.2584

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.025183339	-0.021177839	0.005115365

`lm.morantest(pennlm4, pennwb)`

Global Moran I for regression residuals

data:

```
model: lm(formula = rate ~ smoking, data = penn_sfdf, weights = pop)
weights: pennwb
```

Moran I statistic standard deviate = 2.2477, p-value = 0.0123

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.148285223	-0.014781749	0.005263293

```
lm.morantest(pennlm1, pennws)
```

Global Moran I for regression residuals

data:

```
model: lm(formula = rate ~ 1, data = penn_sfdf)
```

```
weights: pennws
```

Moran I statistic standard deviate = 1.1681, p-value = 0.1214

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.07298525	-0.01515152	0.00569310

```
lm.morantest(pennlm2, pennws)
```

Global Moran I for regression residuals

data:

```
model: lm(formula = rate ~ 1, data = penn_sfdf, weights = pop)
```

```
weights: pennws
```

Moran I statistic standard deviate = 3.304, p-value = 0.0004766

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.236297100	-0.013419831	0.005712493

```
lm.morantest(pennlm3, pennws)
```

Global Moran I for regression residuals

data:

```
model: lm(formula = rate ~ smoking, data = penn_sfdf)
```

```
weights: pennws
```

Moran I statistic standard deviate = 0.69402, p-value = 0.2438

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.030083778	-0.021604759	0.005546848

```
lm.morantest(pennlm4, pennws)
```

Global Moran I for regression residuals

data:

model: `lm(formula = rate ~ smoking, data = penn_sfd, weights = pop)`

weights: pennws

Moran I statistic standard deviate = 2.483, p-value = 0.006514

alternative hypothesis: greater

sample estimates:

Observed Moran I	Expectation	Variance
0.170750634	-0.015142888	0.005605005

Solution

That they are different estimates where the p-values are a lot more statistically significant for the smoking data. Where the population weights gave us a better idea accounting for why we might see larger counts of cancer with areas of higher population.

(f)

Why is it appropriate to adjust the weights of the residuals to be related to the population size of each county? What do these weights imply about the variance associated with each county (see the Details section of the `lm` documentation).

Solution

When you add in the weights option you run a weighted least squares which is minimising the sum of the weights times the error. Thus, it provides more importance to the errors of specific counties with higher population, adjusting for them. It is appropriate since you are more likely to have more cases of cancer simply by having a larger population size. It also means that you are more likely to have less variation within these populations that we want caught in our estimates. It also helps with heteroskedasticity however I am not 100 percent sure this is important to spatial statistics.

(g) [link](#)

Why would you use a row-standardized spatial weights matrix instead of a binary spatial weights matrix?

Solution

You would use a row-standardized spatial weights matrix instead of a binary spatial weights matrix if you believe that there is no “cross contamination” between your different regions. For example in this case you can’t get cancer from someone so you don’t want the extra added weights to that county simply because it has more counties around it.

(h)

Construct one-parameter SAR models using **rate** as the response and **smoking** as a covariate scaling the residual error variances by a constant related to the inverse of the population size of each region and using a row-standardized spatial proximity matrix.

Provide a table with rows for the **Intercept** term, the **smoking** variable, and the ρ parameter. The columns of the table should be the estimated value, the estimated standard error of the estimated value, and the p-value (for testing whether the parameter differs from zero).

Solution

```
# fit one-parameter SAR model with same covariates
ratesar <- spatialreg::spautolm(rate ~ smoking,
                                data = penn_sfdf, listw = pennws, weights = pop)
summary(ratesar)
```

```
Call: spatialreg::spautolm(formula = rate ~ smoking, data = penn_sfdf,
                           listw = pennws, weights = pop)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-5.5493e-04	-1.1229e-04	-3.1714e-05	5.2059e-05	5.0256e-04

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.00039400	0.00011916	3.3066	0.0009445
smoking	0.00190459	0.00048851	3.8988	9.667e-05

Lambda: 0.33286 LR test value: 4.0587 p-value: 0.043946

Numerical Hessian standard error of lambda: 0.15616

Log likelihood: 492.6419

ML residual variance (sigma squared): 0.0021831, (sigma: 0.046723)

Number of observations: 67

Number of parameters estimated: 4

AIC: NA (not available for weighted model)

Parameter	Estimate	Std. Error	p-value
Intercept	0.00039400	0.00011916	0.0009445

Parameter	Estimate	Std. Error	p-value
Smoking	0.00190459	0.00048851	0.00009667
rho	0.33286	0.15616	0.043946

(i)

Is there evidence of residual spatial autocorrelation for the SAR model fit in the previous problem? Why?

Solution

Yes, as we can see from the low p-value of all of our parameters. We know they are very statistically significant in describing the relationship. In the spatial section with rho we notice that it is positive with a very small pvalue which means that there is some statistical significance to the residual spatial autocorrelation.